Intelligent Customer Churn Prediction & Customer Risk Analysis System

Internship Project – AI/ML Development

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Role: AI/ML Intern
Technology Used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

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1. Introduction

Customer churn is an important problem for businesses because losing existing customers can affect their revenue and growth. The main purpose of this project is to use customer data and machine learning to identify customers who are more likely to leave a service.

In this project, I worked on a **Customer Churn Prediction and Risk Analysis System**. The system takes customer-related information such as monthly charges, tenure, usage, login frequency, support tickets and customer satisfaction, and uses this information to predict the possibility of churn.

The project was developed as part of my AI/ML internship to understand the complete machine learning workflow, starting from data preparation and analysis to model training, evaluation and churn probability prediction.

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2. Project Objective

The main objectives of this project were:

- To understand and prepare customer data for machine learning.
- To clean missing and incorrect data.
- To perform exploratory data analysis (EDA).
- To identify patterns related to customer churn.

- To prepare features for machine learning.
- To train different classification models.
- To compare the performance of the models.
- To identify important features affecting the model's predictions.
- To calculate the probability of customer churn.
- To divide customers into different risk levels.

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3. Dataset Description

The dataset used in this project contains **10,000 customer records** and **18 original columns**.

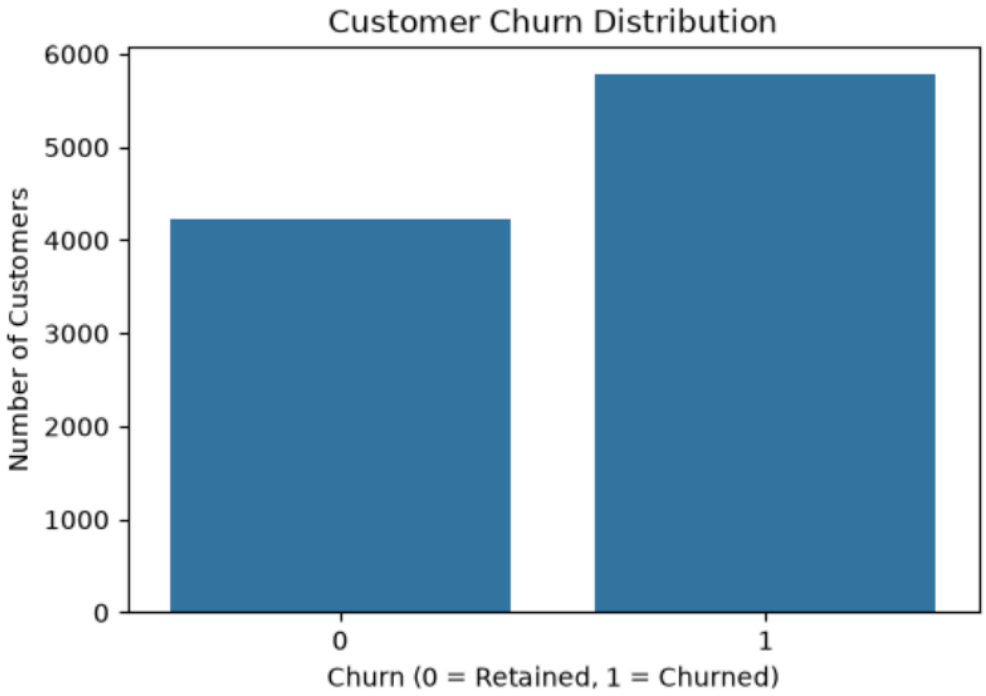
Some of the important columns are:

Feature	**Description**
Customer_ID	Unique customer identifier
Age	Age of the customer
Gender	Customer gender
Location	Customer location
Tenure_Months	Number of months the customer has been with the service
Monthly_Charges	Monthly amount paid by the customer
Total_Charges	Total amount charged
Subscription_Type	Type of subscription
Payment_Method	Method used for payment
Login_Frequency	Frequency of customer logins
Support_Tickets	Number of support tickets
Complaints	Number of complaints

Last_Login_Days	Days since the customer's last login
Usage_Hours	Service usage hours
Customer_Satisfaction	Customer satisfaction score
Churn	Target variable

For the **Churn** column:

- `0` = Customer retained
- `1` = Customer churned



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4. Data Preparation

Before training the models, I checked the dataset for missing values, duplicate records and invalid values.

Initially, some missing values were present in:

- Monthly Charges
- Usage Hours
- Customer Satisfaction

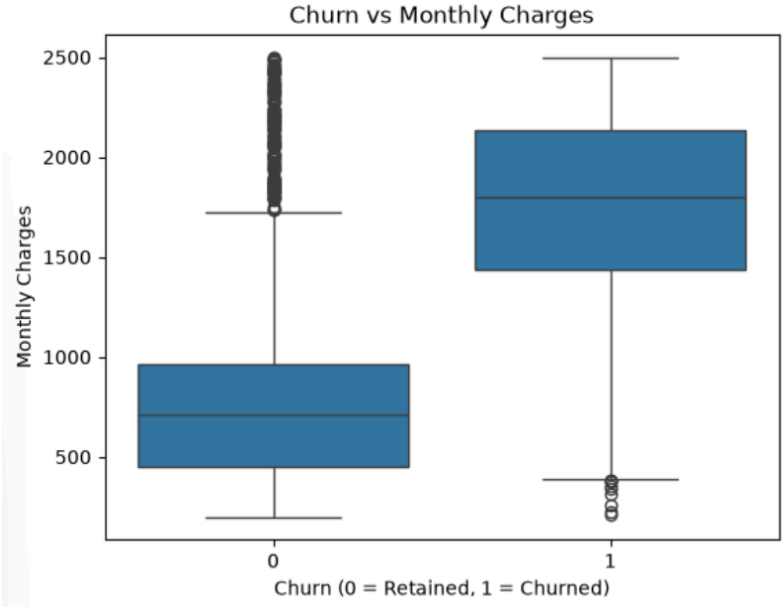
These missing values were handled using the **median value of the training data**.

I also checked for duplicate customer IDs and invalid values. No duplicate Customer IDs or invalid values were found in the final dataset.

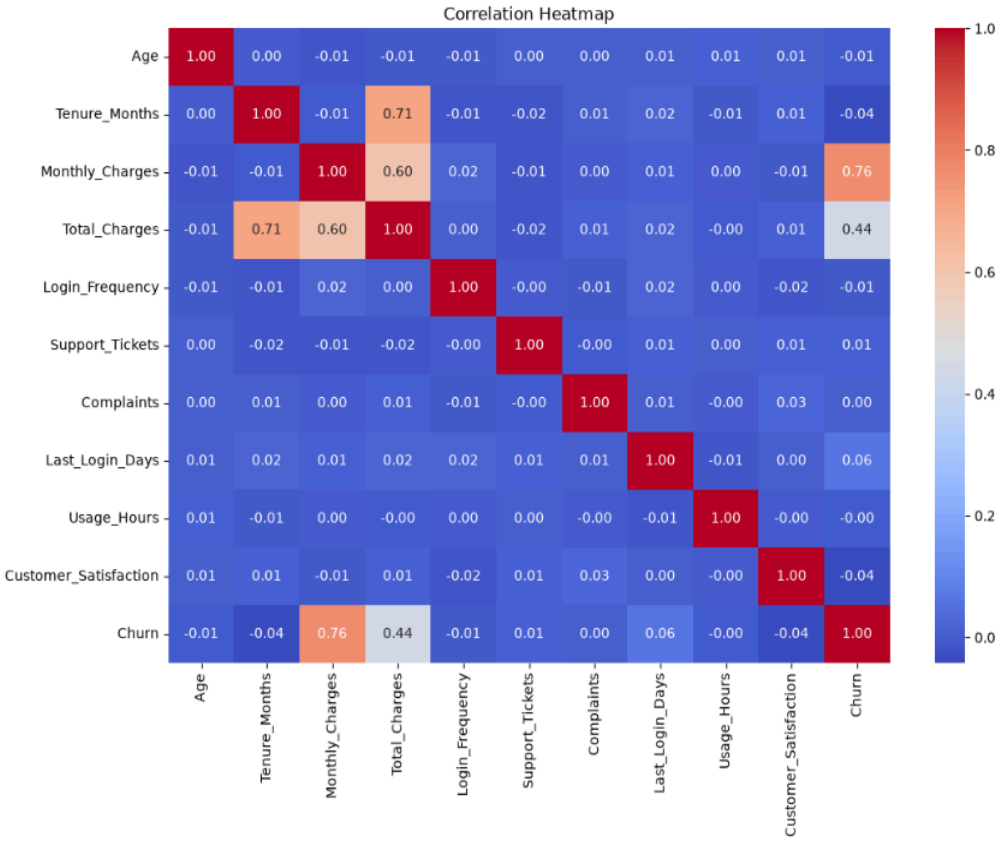
After preprocessing, the data was ready for machine learning.

5.Exploratory Data Analysis

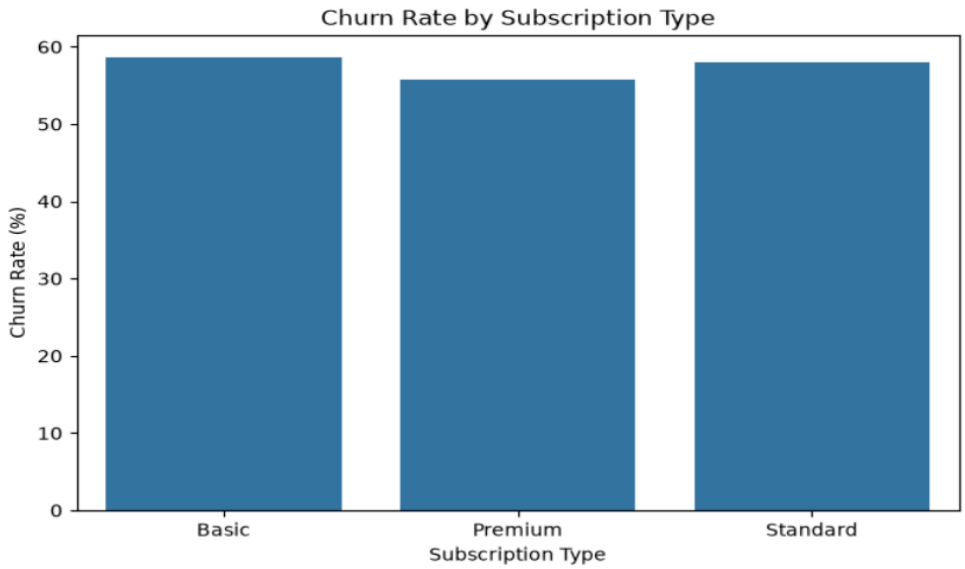
I performed exploratory data analysis to understand the relationship between customer attributes and churn. Different numerical and categorical features were compared with the churn variable to identify noticeable patterns in the dataset.



The chart shows that customers who churned generally had higher monthly charges compared with retained customers.



The correlation heatmap was used to understand the relationships between numerical features and the churn variable. Monthly Charges shows the strongest positive correlation with Churn among the numerical features.



The chart compares the churn rate across different subscription types. The differences between the subscription categories can be observed from the chart.

6. Model Development

After preparing the data, I trained three different machine learning classification models to predict whether a customer is likely to churn.

The models used were:

6.1 Logistic Regression

Logistic Regression was used as the first model because it is a commonly used classification algorithm and provides a good baseline for comparison.

The model was trained using the prepared customer data and was used to predict whether a customer would churn (`1`) or remain with the service (`0`).

6.2 Decision Tree

The second model used was Decision Tree.

A Decision Tree makes predictions by creating a series of conditions based on the input features. It was used to check how well a tree-based model could classify customers into churn and non-churn categories.

6.3 Random Forest

The third model was Random Forest. Random Forest combines multiple decision trees and uses their combined predictions. It was trained to identify patterns in customer behaviour and predict churn.

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7. Model Evaluation

After training the models, I evaluated them using **Accuracy, Precision, Recall and F1-Score**.

The results obtained were:

Model	Accuracy	Precision	Recall	F1-Score

Logistic Regression	91.35%	91.62%	93.60 %	92.60%
Decision Tree	85.90%	88.47%	86.94 %	87.70%
Random Forest	**92.05%**	91.58%	**94.98 %**	**93.25%**

From the results, Random Forest gave the highest accuracy and recall among the three models tested. Its F1-score was also the highest in my experiments.Since identifying customers who may churn is an important part of this project, I selected the **Random Forest model for the further churn probability and risk analysis steps**.

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8. Feature Importance

After training the Random Forest model, I checked the feature importance values to understand which customer attributes contributed more to the model's predictions.

The top features were:

Feature	Importance
Monthly_Charges	0.5527
Total_Charges	0.1573
Tenure_Months	0.0644
Usage_Hours	0.0310
Last_Login_Days	0.0296
Customer_Satisfaction	0.0273

Age	0.0261
Login_Frequency	0.0248
Support_Tickets	0.0121
Complaints	0.0096

Monthly Charges had the highest feature importance in the Random Forest model, followed by **Total Charges** and **Tenure Months**.

These values show which features were most useful to the trained model when making its predictions. They should not be interpreted as proof that these factors directly cause customers to churn.

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9. Churn Probability Prediction

After selecting the Random Forest model, I used `predict_proba()` to calculate the probability of churn for each customer.

Instead of giving only a `0` or `1` prediction, the model provides a probability value between **0 and 1**.

For example:

- `0.97` → 97% predicted churn probability
- `0.39` → 39% predicted churn probability
- `0.05` → 5% predicted churn probability

This gives more information about the level of risk for each customer.

10. Customer Risk Classification

To make the probability easier to understand from a business point of view, I divided customers into three risk levels:

Churn Probability	Risk Level
0% – 30%	Low
31% – 70%	Medium
71% – 100%	High

For example, in my prediction results:

- Customer `CUST10001` → **97%** → **High Risk**
- Customer `CUST10002` → **39%** → **Medium Risk**
- Customer `CUST10003` → **5%** → **Low Risk**
- Customer `CUST10005` → **96%** → **High Risk**

This makes the model output easier to understand and can help identify customers who may need attention.

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11. Project Workflow

The project was completed by following a step-by-step machine learning workflow. The main stages were:

Customer Data → Data Cleaning → Data Analysis → Feature Preparation → Train/Test Split → Model Training → Model Evaluation → Churn Probability → Risk Classification

Each stage helped prepare the data for the next step and finally convert the model predictions into customer risk information.

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12. Tools and Technologies Used-

The following tools and technologies were used during the project:

- **Python** – Main programming language used for the project.
- **Pandas** – Used for loading, cleaning and analysing the dataset.
- **NumPy** – Used for numerical operations.
- **Matplotlib** – Used for creating visualizations.
- **Seaborn** – Used for analysing and visualizing patterns in the data.
- **Scikit-learn** – Used for data splitting, model training and evaluation.
- **Jupyter Notebook** – Used to develop and test the machine learning workflow.

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13. Key Findings

During the analysis and model development, I observed some important patterns in the customer data.

- The dataset contained **10,000 customer records**.
- The churn column was used as the target variable.
- **Monthly Charges** was the most important feature in the Random Forest model.
- **Total Charges** and **Tenure** were also among the important features.
- Random Forest achieved an accuracy of **92.05%** on the test data.
- The Random Forest model achieved a recall of **94.98%**.
- The model can provide a **churn probability** instead of only giving a yes/no prediction.
- Customers can be grouped into **Low, Medium and High risk levels** based on their predicted churn probability.

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14. Business Use of the Model

The main practical use of this project is to help businesses identify customers who may be at a higher risk of leaving.

For example, if a customer receives a high churn probability, the business could review that customer's usage, charges, satisfaction and support history and decide whether any retention action is appropriate.

The model can therefore be used as a **decision-support tool** rather than replacing human decision-making.

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15. Conclusion

This project helped me understand and implement the complete machine learning workflow for a customer churn problem.

I started with customer data, performed data cleaning and analysis, prepared the data for machine learning, trained different classification models and compared their performance.

Among the tested models, **Random Forest performed best on the evaluation metrics used in this project**. I then used the model to generate churn probabilities and classify customers into different risk levels.

Overall, this project gave me practical experience in **data preprocessing, exploratory data analysis, machine learning, model evaluation and business-oriented prediction**.

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16. Future Scope

The project can be improved further by adding:

- A **Streamlit dashboard** for an interactive user interface.
- A customer search option to view individual risk details.
- Automatic generation of customer risk reports.
- A recommendation system for different customer risk levels.
- Regular retraining of the model when new customer data becomes available.
- Database integration using **SQLite or MySQL**.
- Model monitoring to check performance on new data.

17. Final Project Outcome

The final system is able to take customer information, process the data, use a trained machine learning model to estimate churn probability and classify customers according to their risk level.

The project demonstrates how machine learning can be used to convert customer data into useful information that can support customer retention activities.